

What Do Text Classifiers Learn?

Ablation, Masking and Robustness Analysis on AG News

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Problem

What do text classifiers learn? What do they rely on?

- actual context?
- shallow cues or shortcut?

Accuracy alone doesn't explain everything

Research Questions

RQ1

Title vs Description vs Title+Desc — input information content

RQ2

Classical ML vs Neural CNN vs Pretrained Transformer

RQ3

How much do models depend on obvious topic keywords?

RQ4

How robust are models against noise / input perturbation?

RQ5

Are confidence scores trustworthy, especially when wrong?

Dataset & Experimental Setup

AG News (4-class topic classification)

- Classes: World / Sports / Business / Sci-Tech
- Each sample: title + description + label
- Train 108K / Val 12K / Test 7.6K
- Stratified split, seed=42

Evaluation

- Accuracy / Macro-F1 (main metric)
- Per-class Precision / Recall / F1
- Confusion matrix
- Expected Calibration Error (ECE)
- Accuracy drop Δ from baseline

GPU: A100-80GB (Colab Pro+) · total training time $\approx$ 90 min · 3 models $\times$ 3 input modes = 9 configs

ML Models

**TF-IDF
+ Logistic
Regression**

Fast and
interpretable baseline

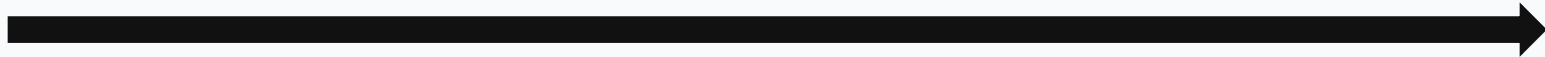
TextCNN

Learns local phrases

DistilBERT

Uses pretrained
contextual knowledge

Model complexity increases



Approach — Five Experiments

1

Input ablation

title vs description vs title + description

2

Robustness analysis

typo / source-marker removal / text norm. / NER masking

3

Calibration & HC-wrong

ECE · reliability diagrams · high-confidence wrong cases

4

Masking analysis

random vs keyword vs attribution-based masking

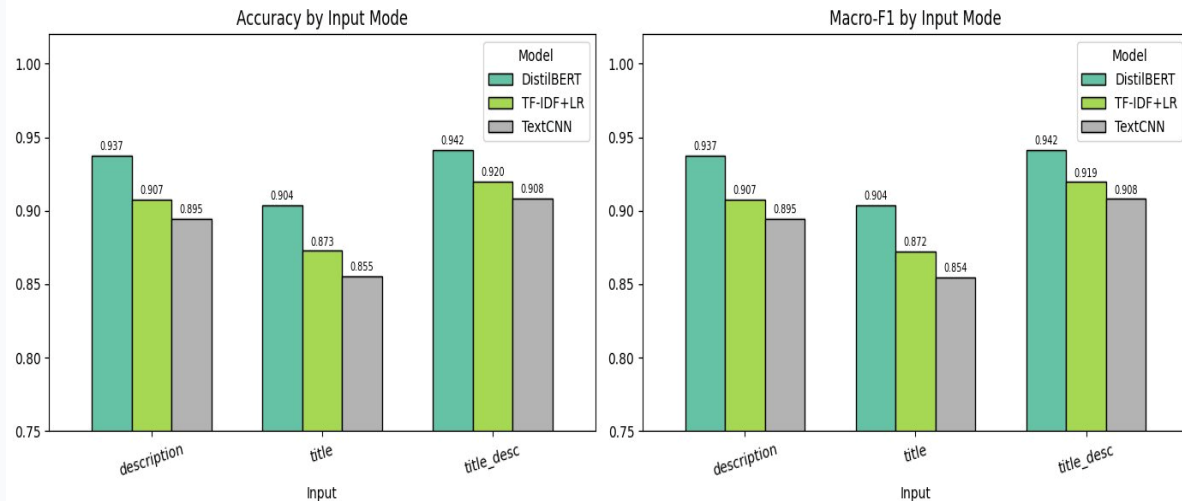
5

Learning curve

sample efficiency across 1K → 108K training sizes

Exp 1 — Input Ablation

Input Ablation — All Models

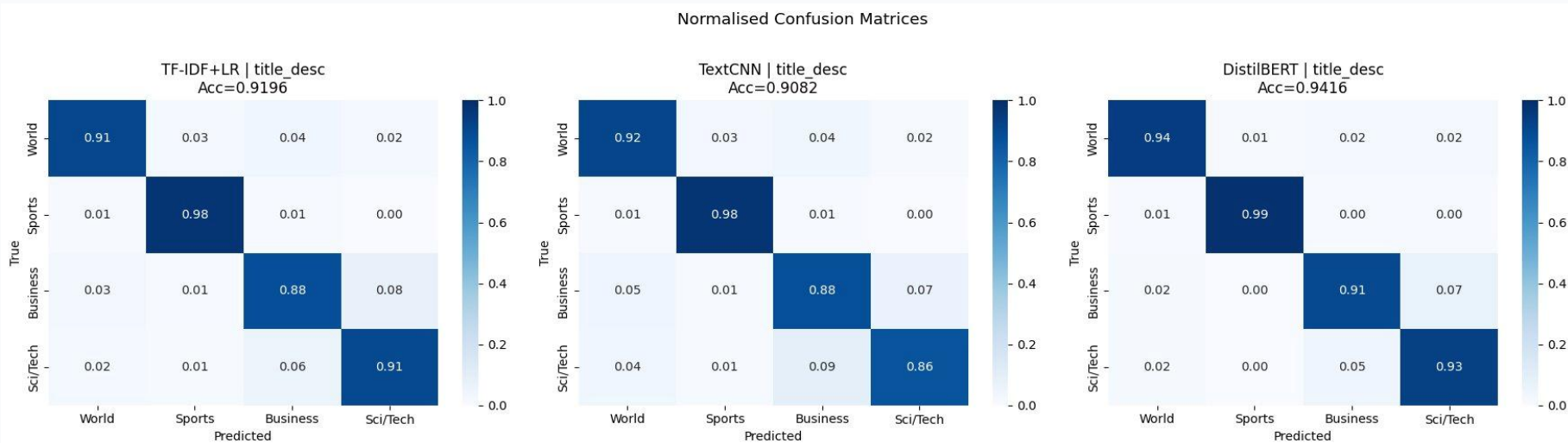


Key findings

- title_desc > description > title across all 3 models
- description alone is ~3%p stronger than title alone
- Adding title gives ~1%p extra → complementary info
- Model order invariant: BERT > LR > CNN

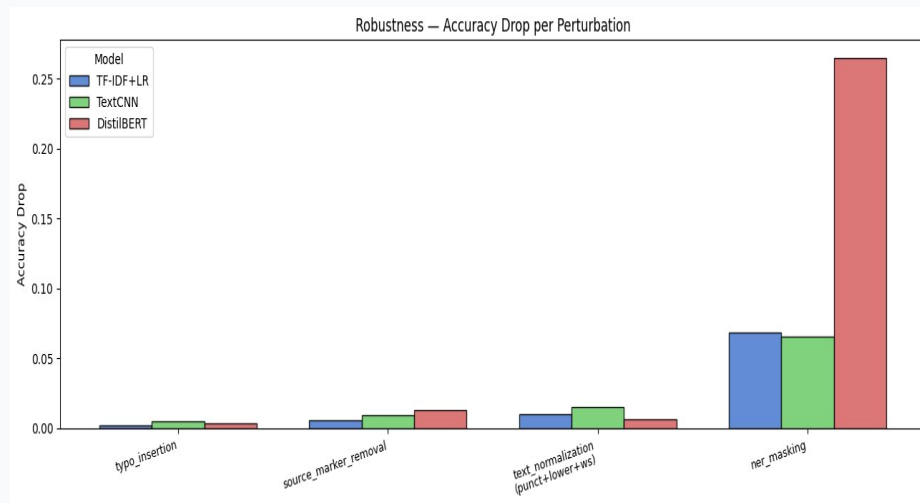
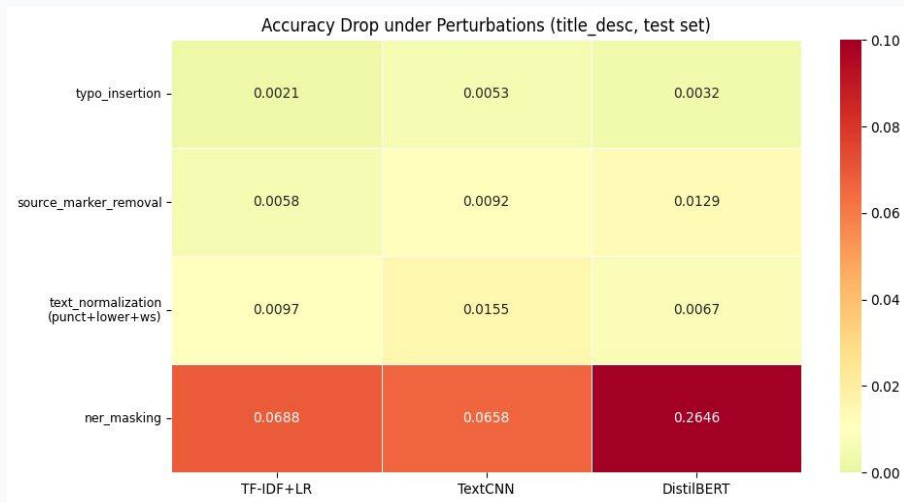
Best: DistilBERT (title_desc) — Acc 0.9416 / Macro-F1 0.9416

Per-class Behavior — Confusion Matrices



- Sports is the easiest class for every model (≥ 0.98)
- Business $\leftrightarrow$ Sci/Tech is the dominant confusion pair (7–9% each way)
- DistilBERT lifts the two hardest classes the most (World 0.94, Sci/Tech 0.93)

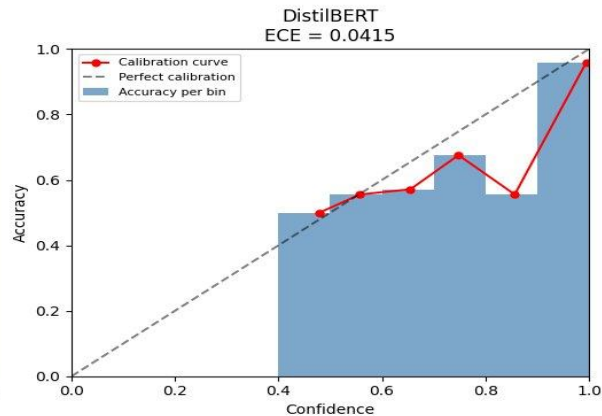
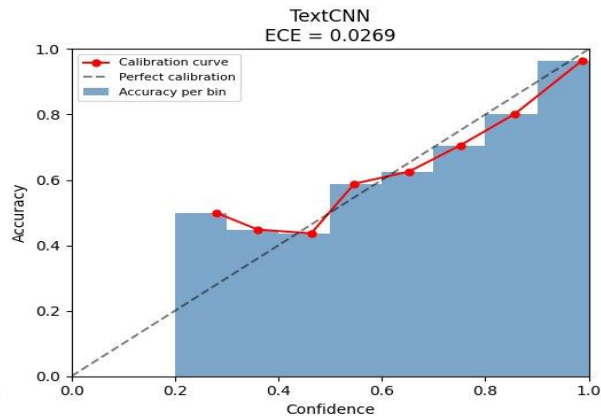
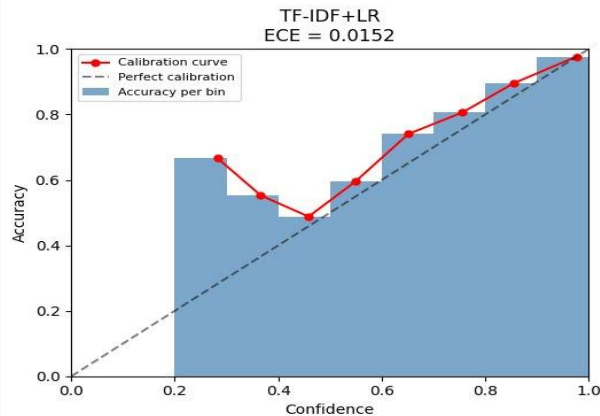
Exp 2 — Robustness to Perturbations



- Typo / source-marker / text-norm.: all models robust ($\leq 1.5\%$ drop)
- **NER masking: DistilBERT -26.5% , others $\approx 6.5\%$ — BERT is 4× more fragile**
- **Most accurate model is NOT the most robust**

Exp 3 — Calibration (Reliability Diagrams)

Reliability Diagrams (title_desc, test set)



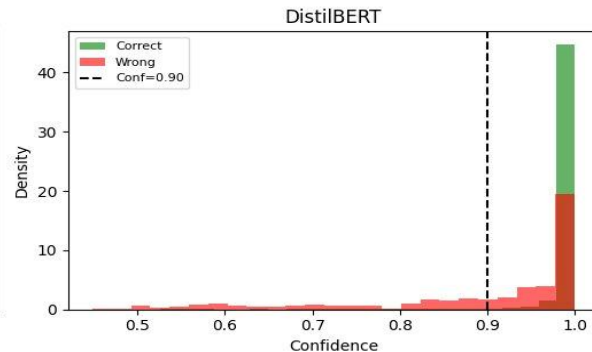
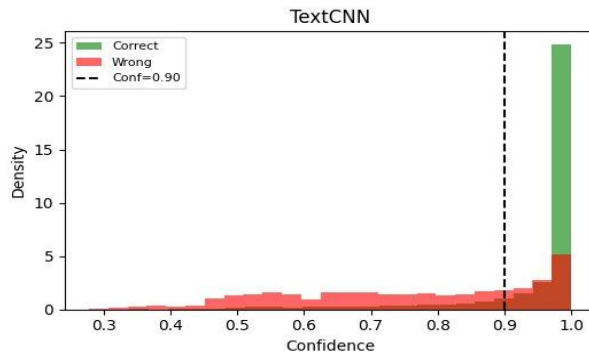
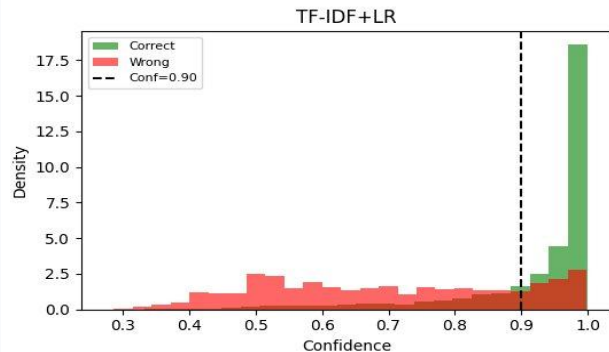
TF-IDF + LR
ECE = 0.0152
best calibrated

TextCNN
ECE = 0.0269
moderate

DistilBERT
ECE = 0.0415
over-confident

Confidence Distribution & HC-Wrong Cases

Confidence Distribution: Correct vs Wrong



TF-IDF + LR

1.67%

HC-wrong 127 / 7,600

TextCNN

2.84%

HC-wrong 216 / 7,600

DistilBERT

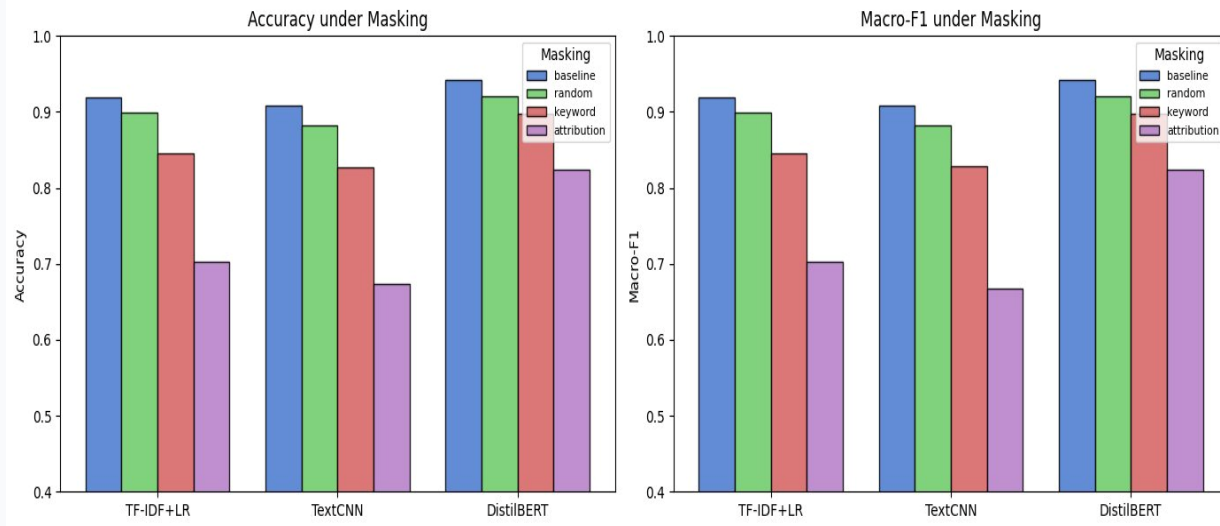
3.93%

HC-wrong 299 / 7,600

DistilBERT confidently wrong 2.4× more often than TF-IDF — accuracy & trustworthy confidence are inversely related

Exp 4 — Masking Analysis

Masking Experiments — 3 Techniques x 3 Models



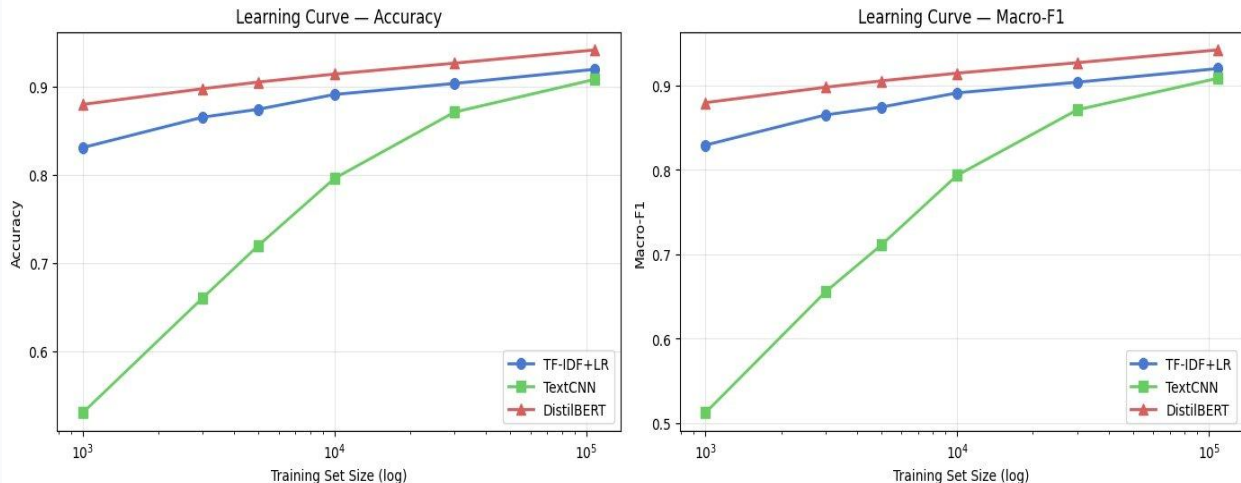
Three masking conditions

- **Random 30% (control)**
→ only ~2%p drop, all models
- **Top-100 class keywords**
→ LR/CNN $\approx 7-8\%$ p, BERT only 4.4%p
- **Top-5 attribution words**
→ LR/CNN drop $\approx 22\%$ p; BERT only 12%p

DistilBERT spreads its reliance across many cues; LR/CNN are dominated by a small set of decisive words

Exp 5 — Learning Curve

Learning Curves (title_desc, test set)



@ 1,000 samples

DistilBERT

0.880

96% of LR full-data perf.

TF-IDF + LR

0.831

stable hand-crafted feat.

TextCNN

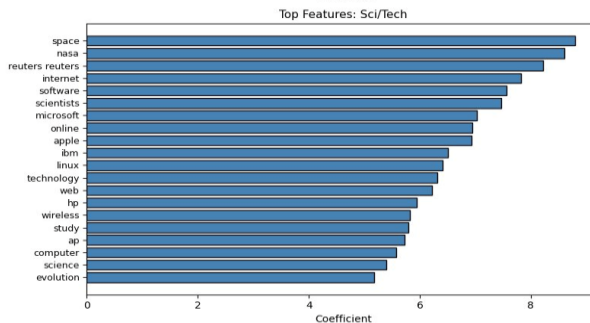
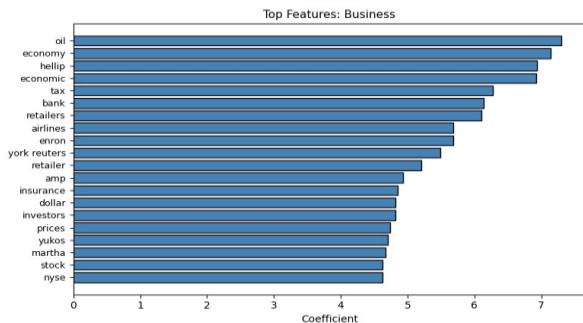
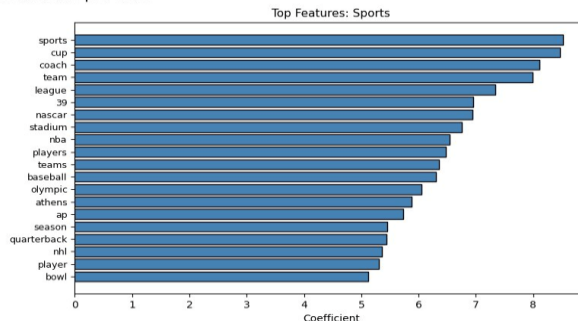
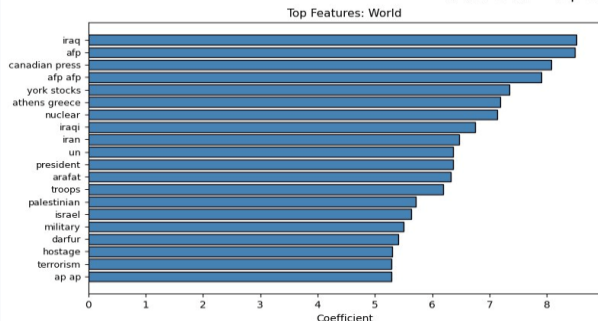
0.530

cold-start collapse

Curves saturate around 30K — AG News is a relatively shallow task

Interpretation — TF-IDF + LR Coefficients

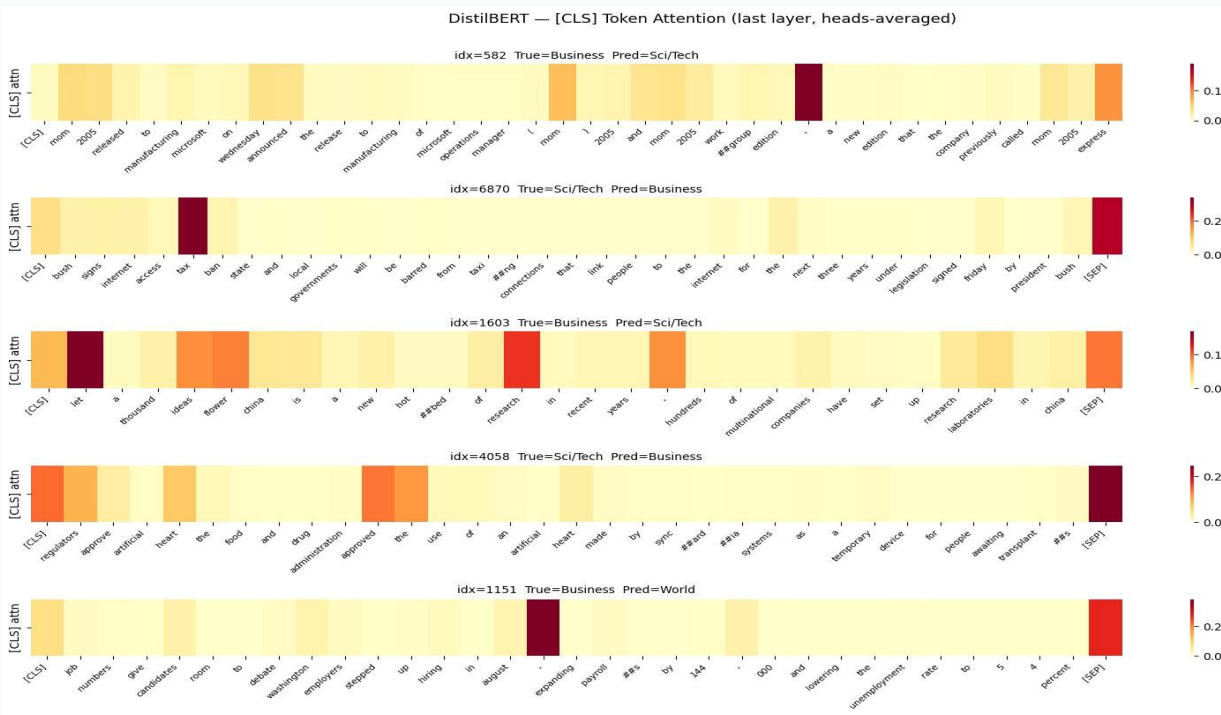
TF-IDF + LR — Top 20 Coefficients per Class



What LR learned

- Meaningful keywords dominate (iraq, nasa, oil, cup...)
- But spurious source tokens also rank high (afp, reuters reuters)
- Bigrams like "york stocks" carry real signal
- Tokenization artifacts visible (e.g. "39")

Interpretation — DistilBERT [CLS] Attention



What BERT attends to

- [CLS] / [SEP] dominate (architectural)
- Content attention falls on decisive keywords (manufacturing microsoft, tax ban, regulators heart)
- **Direct visual evidence of shortcut learning when wrong**

Answers to the Five Research Questions

RQ1

desc > title by ~3%p; combining gives +1%p (complementary)

RQ2

DistilBERT > TF-IDF+LR > TextCNN (in accuracy; ordering flips for calibration / robustness)

RQ3

All depend on keywords — DistilBERT spreads reliance the most (−4.4 vs −7–8%p)

RQ4

Robust to typo / source / norm., but DistilBERT collapses under NER masking (−26.5%p)

RQ5

LR best calibrated (ECE 0.015); BERT over-confident (ECE 0.041) and most confidently wrong

Trade-off Matrix — No Single Best Model

Criterion	1st	2nd	3rd
Absolute accuracy	DistilBERT	TF-IDF + LR	TextCNN
Sample efficiency	DistilBERT	TF-IDF + LR	TextCNN
Calibration	TF-IDF + LR	TextCNN	DistilBERT
Robustness to noise	TF-IDF + LR	TextCNN	DistilBERT
Interpretability	TF-IDF + LR	TextCNN	DistilBERT
Inference speed	TF-IDF + LR	TextCNN	DistilBERT

Choose the model based on what your deployment values most

Conclusion — Key Takeaways

1 All models exploit shortcuts

but DistilBERT spreads reliance the widest —
most resistant to single-keyword removal

2 BERT's strength is its weakness

entity-centric representations collapse
-26.5%p when entities are masked

3 Accuracy $\neq$ Calibration

the most accurate model gives the least
trustworthy probabilities

4 TextCNN is sample-hungry

random-init embeddings cannot compete with
TF-IDF's hand-crafted features on this task